

2nd Mar 2026 / DA6360

⇒ (contd. - Basics)

$$M_{(T_2 \circ T_1)} = M_{T_2} \cdot M_{T_1}$$

$$M_{T_1} = \begin{bmatrix} 1 & 0 & -3 \\ 2 & 2 & -1 \\ -1 & 5 & 2 \end{bmatrix}$$

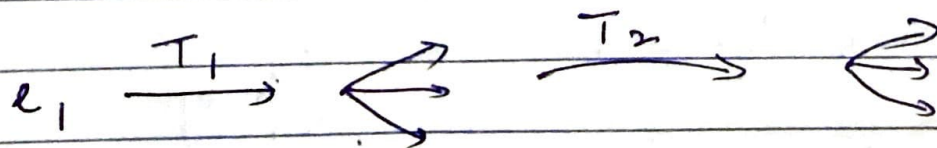
$$M_{T_2} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix}$$

$$M_{(T_2 \circ T_1)} = M_{T_2} \cdot M_{T_1} = \begin{bmatrix} + & 2 & 2 \\ - & - & 3 \\ - & - & - \end{bmatrix}$$

std. procedure
(i, j)

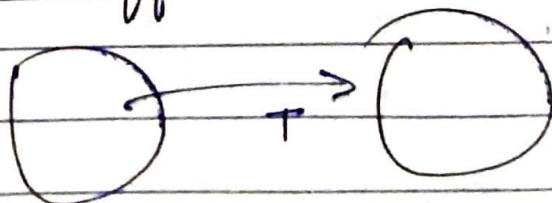
first column of $M_{T_2 \circ T_1}$,

$$1 \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + -1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$



⇒ Matrix Representaⁿ of Same Transformaⁿ
in a diff. basis

same basis
on both
domain
& range



std. basis = M

Basis $V = M_v$

how to present M_v in M & $V = \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix}$

$$\begin{aligned} - \quad x &\in \mathbb{R}^d \text{ (std. basis)} \rightarrow Mx \\ x_v &\in \mathbb{R}^d \text{ (basis } V) \rightarrow M_v x_v \end{aligned}$$

$$V x_v = x \Rightarrow x_v = V^{-1} x$$

$$\begin{array}{ccccc} x_v & \xrightarrow[\text{basis}]{\text{convert to std. basis}} & V x_v & \xrightarrow{T} & M V x_v \rightarrow V^{-1} M V x_v \\ \text{(basis } v) & & & & \text{(std. basis)} \end{array}$$

- if V is invertible, then M & $V^{-1} M V$ are similar matrices. They represent the same transformation in diff. basis.

⇒ Real Symmetric Matrices

- any matrix for $H = H^T$ are similar to a diagonal matrix, & has orthonormal basis.

$$D = V^T H V$$

$$H = V D V^T$$

(project scaling & projecting back)

invariant

$D = \begin{bmatrix} 10 & 0 \\ 0 & 0.1 \end{bmatrix}$, takes a pt. & scales the 1st coordinate by 10 times & second coord. by 0.1.

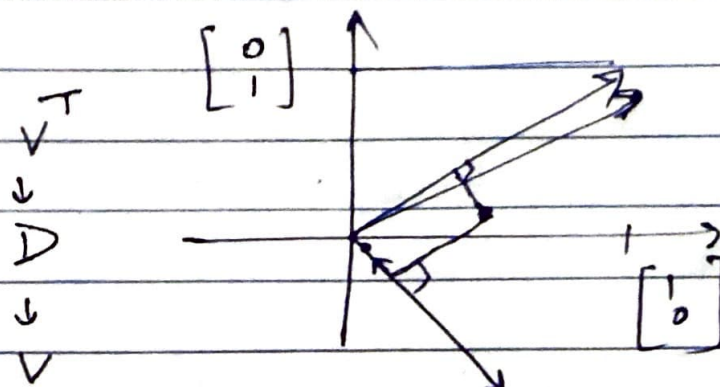
we are only considering positive semi-definite matrices. $x^T H x > 0$

$$\begin{bmatrix} \frac{10}{\sqrt{2}} & \frac{0.1}{\sqrt{2}} \\ \frac{10}{\sqrt{2}} & -\frac{0.1}{\sqrt{2}} \end{bmatrix} \rightarrow \begin{bmatrix} \frac{10.1}{2} & \frac{9.9}{2} \\ \frac{9.9}{2} & \frac{10.1}{2} \end{bmatrix} \quad x \text{ vector}$$

$$H = V D V^T$$

$$= \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}}_V \underbrace{\begin{bmatrix} 10 & 0 \\ 0 & 0.1 \end{bmatrix}}_D \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}}_{V^T} V^T \begin{bmatrix} x \\ x \end{bmatrix}$$

↪ "project"



it's just scaling
in ortho-
normal basis
& then just
project.

$\langle u_i, x \rangle$ is a comp. of x in the new basis.

- Generalized Quadratic Norm :

$$\begin{aligned} x^T H x &> 0 \quad (H \text{ is sym. positive definite}) \\ \|x\|_H &= \sqrt{x^T H x} \\ &= \sqrt{x^T V D V^T x} \\ &= \sqrt{x_v^T D x_v} \quad \xrightarrow{x_v} \\ \|x\|_2^2 &= \|x_v\|_2^2 = (V^T x)^T (V^T x) \\ &= x^T V V^T x \\ &= x^T x \end{aligned}$$

→ Unconstrained Optimization in $\mathbb{R}^d$

$$\min_{x \in \mathbb{R}^d} f(x)$$

Cutting Plane & Ellipsoidal Methods → ① & ②

① Slice the space
consider a following problem

$$f(x) = \sum_{i=1}^{10^6} (x_i - 0.0001)^2$$

$$x_* = (0.0001) \quad \text{in all the coordinates}$$

let's say we cut one coordinate at a time starting from

$$x_0 = [0, 1] \times [0, 1] \dots$$

10^6 times

$$x_1 = [0, 1/2] \times [0, 1] \times [0, 1] \dots$$

$$x_2 = [0, 1/2] \times [0, 1/2] \times [0, 1] \dots$$

$$x_{1000} = [0, 1/2] \dots \quad \text{1000 times} \quad [0, 1] \dots \quad \text{10}^3 \text{ times}$$

$$\text{energy} = \sqrt{\left(\frac{1}{2} - 0.0001\right)^2 + \left(\frac{1}{2} - 0.0001\right)^2 + \dots + \left(1 - 0.0001\right)^2 + \dots}$$

(1000 times) (1000 times)

$$\approx \sqrt{10^6} = 10^3$$

② Shrink the space
each dim. len is l .

$$\text{Vol.} = l \times l \dots = l^d$$

say I cut one dim. at a time.

$$\text{Vol.}_{k+1} = \frac{l}{2} \times l \dots = \frac{l^d}{2}$$

$$= \left(\left(\frac{1}{2} \right)^{1/d} l \right)^d$$

$$d = 1000, \left(\frac{1}{2} \right)^{1/1000} = 0.999. \dots$$

not much shrinkage.

③ discrete based (gradient descent)

$$l_t = (1 - \alpha)^t l_0, \quad \alpha = 0.1$$
$$= (0.9)^t l_0$$

- picking up more useful defⁿ that'll help us to think abt. optimisaⁿ in $\mathbb{R}^d$

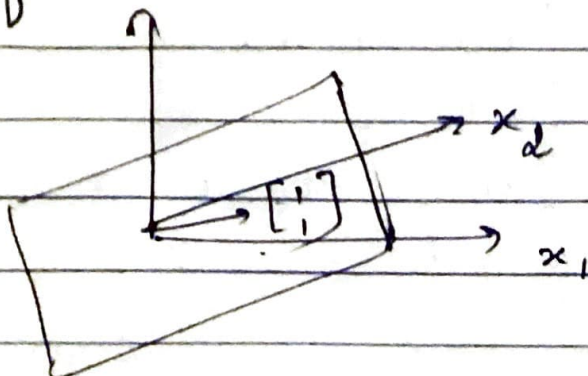
- Graph of a funcⁿ

$$G(f) = \{ (x_1, \dots, x_d, y) \in \mathbb{R}^{d+1} \}$$

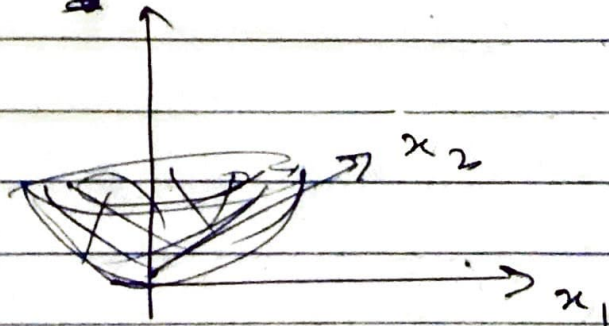
$$y = f(x_1, \dots, x_d)$$

(not like $|x|$)

$$f(x) = x_1 + x_2$$



$$f(x) = x_1^2 + x_2^2$$

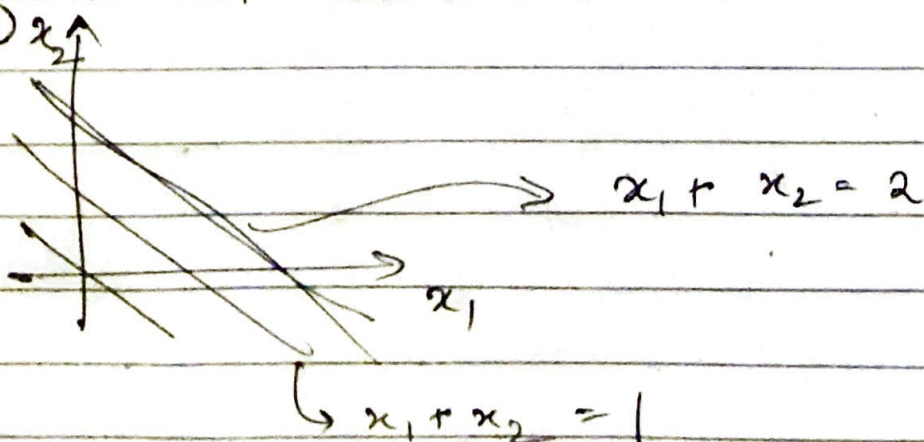


- Level Surface or Contour

$$\text{contour}(f, \beta) = \{ (x_1, \dots, x_d) \in \mathbb{R}^d \}$$

$$\beta = f(x_1, \dots, x_d)$$

$$f(x) = x_1 + x_2$$



$$f(x) = x_1^2 + x_2^2$$

